

Two-Step Equations

AQA · KS3 Year 8

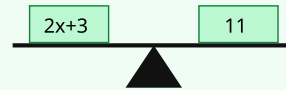
Answer each question in your workbook. Show your working.

VISUAL EXAMPLE

$$2x + 3 = 11$$

$$-3 \text{ both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



An equation balances: do the same to both sides to keep it level.

COMMON MISTAKES

Dividing before undoing the + or -.

Not doing each step to both sides.

$2x = 8$ gives $x = 4$, not $2x = 4$.

Losing a minus sign with negative constants.

METHOD

- 1 Spot the letter term and the + / - number
- 2 Undo the + or - first (both sides)
- 3 Now it is a one-step equation
- 4 Undo the $\times$ or $\div$ (both sides)
- 5 Check by substituting

WORKED EXAMPLE

Solve $2x + 3 = 11$.

Subtract 3: $2x = 8$. Divide by 2: $x = 4$.

Check: $2 \times 4 + 3 = 11$

VISUAL AID

Balance scale: $2x + 3$ equals 11. Remove 3 from both sides to get $2x = 8$, then halve both sides to get $x = 4$.

1. (2 marks)

Solve: a) $2x + 1 = 7$ b) $3x + 4 = 19$

2. (2 marks)

Solve: a) $5x - 2 = 13$ b) $4x - 7 = 9$

3. (2 marks)

Solve: a) $x \div 2 + 3 = 8$ b) $x \div 3 - 1 = 4$

4. (3 marks)

Solve: a) $2x + 9 = 3$ b) $5x + 14 = 4$ c) $3x - 5 = -11$

5. (3 marks)

Solve: a) $7 = 2x + 1$ b) $10 = 4x - 6$ c) $(x + 5) \div 2 = 6$

6. (3 marks)

Form and solve: a) Taxi £3 + £2/mile = £17 — how many miles?
b) Number $\times 3$ then $- 5 = 16$.

7. (3 marks)

Rectangle length $(2x+1)$, width 4, perimeter 30 cm. a) Equation.
b) Find x . c) Length.

8. (5 marks)

Repair: £15 call-out + £c/hour. 3-hour job = £60. a) Find c . b) Cost of 5 hours. c) £75 job — how many hours?